Homework 3

Problem 1. (Exercise 2.3.4 (c)) Convert regular expression $00(0+1)^*$ to an NFA with ϵ-transitions using Thompson's method.

Solution.

Problem 2. (Exercise 2.3.6) Let $A = (Q, \Sigma, \delta, q_0, \{q_f\})$ be an ϵ-NFA such that there are no transitions into q_0 and no transitions out of q_f. Describe the language accepted by each of the following modifications of A, in terms of $L = L(A)$:

(a) The automaton constructed from A by adding an ϵ-transition from q_f to q_0.

(b) The automaton constructed from A by adding an ϵ-transition from q_0 to every state reachable from q_0 (along a path whose labels may include symbols of Σ as well as ϵ).

(c) The automaton constructed from A by adding an ϵ-transition to q_f from every state that can reach q_f along some path.

(d) The automaton constructed from A by doing both (b) and (c).

Solution.

Problem 3. (Exercise 3.2.3) Convert the following DFA to a regular expression, using the state-elimination technique of Section 3.2.2.

	0	1
$\rightarrow *p$	s	p
q	p	s
r	r	q
s	q	r

Solution.

Problem 4. Prove or disprove each of the following statements about regular expressions.

(a) (Exercise 3.4.1 (g)) $(\epsilon + R)^* = R^*$.

(b) (Exercise 3.4.2 (b)) $(RS + R)^*R = R(SR + R)^*$.

(c) (Exercise 3.4.2 (d)) $(R + S)^*S = (R^*S)^*$.

Solution.

Problem 5. Construct a DFA for each of the regular languages below:

(a) All strings over $\{a, b\}$ that have and only have odd number of substrings ab.

(b) All strings over $\{a, b\}$ that contain ab but not bb as substring.

Solution.

Problem 6. Give a regular expression for each of the regular language below:

(a) $\{xwx^R \mid x, w \in (a+b)^+\}$.

(b) $\{w \mid w \in \{0,1\}^*, w$ has at least three 1's and the third from the last position is a $1\}$.

Solution.